Roll No. : ………………… ………….

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech** Semester: V^{th}

Name of the Course : System Software Course Code: TCS 501

Time : 90 Minutes **Maximum Marks : 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1 (10 Marks) CO1

(a) What is the requirement for software Licenses? Explain. What are the different types of software licenses?

OR

(b) "Assembler is categorized as a system software". Is this statement true or false? Justify your answer.

Q2 (10 Marks) CO3

(a) What is the significance of the relocatable object program? Write the relocatable object program of the following SIC/XE assembly program by relacing XXXX with the appropriate address.

```
MID1     START      XXXX
FIRST    LDX (04)   #0
         LDA (00)   #0
         +LDB (68)  #TABLE2
         BASE       TABLE2
LOOP     ADD (18)   TABLE,X
         ADD (18)   TABLE2,X
         TIX (2C)   COUNT
         JLT (38)   LOOP
         +STA (14)  TOTAL
         RSUB (4C)
COUNT    RESW       1
TABLE    RESW       2560
TABLE2   RESW       2560
TOTAL    RESW       3
         END        FIRST
```

OR

(b) Write the object program of the following SIC/XE Source code

MID2	START	7000
INLOOP	TD(E0)	=X'F1'
	JEQ(30)	INLOOP
	RD(D8)	=X'F1'
	STCH (54)	DATA
OUTLP	TD (E0)	=X'05'
	JEQ (30)	OUTLP
	LDCH(50)	DATA
	WD(DC)	=X'05'
DATA	RESB	1
	END	

Q3 (10 Marks) CO3

(a) Considering following SIC program, write the object program.

MID3	START	0
LABI	LDX (04)	ZERO
	STA (0C)	LENGTH
COPY	LDA (00)	ARRAY1,X
	MUL (20)	NUM1
	DIV (24)	NUM2
	STA (0C)	ARRAY2,X
	TIX (2C)	LENGTH
	JLT (38)	COPY
LENGTH	WORD	1366
ARRAY1	RESW	1366
ARRAY2	RESW	1366
NUM1	WORD	30
NUM2	WORD	7
	END	LABI

OR

(b) Justify with proper reason, the requirement of define and refer records in case of program with multiple control section? Write the syntax for define and refer records.

Q4 (10 Marks) CO2

(a)
1. Disassemble the following sic object program (assume the meaningful name for the labels) opcodes: LDA -00, STA-0C, LDCH-54, STCH-54. Hexcode for ascii character Z is 5A.

```
HEXAM4 001000000014
T0010000C00100F0C100C501012541013
T00100F040000055A
E001000
```

OR

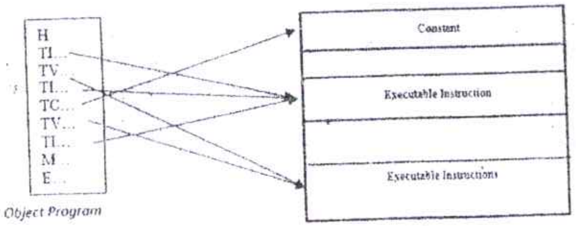
(b) Write short note on expressions. What are the different types of expressions available in assembler. Explain each.

What would be the difference between the following sequences of statements

i. LDA LENGTH
   SUB #1

ii. LDA LENGTH - 1

Q5	(10 Marks)	
(a)	Suppose that a relocatable SIC/XE program is to be assembled in three different parts. One part contains the assembled instructions of the program (LDA, JSUB, etc.) Another part contains the data variables used in the program (which are defined by RESW, BYTE, RESB, and WORD). The third part contains data constants (which are defined by a new assembler directive named CONST).	CO2



Object Program

Memory

In the object program, the assembled instructions are contained in type TI records, the variable in type TV records, and the constants in type TC records. (These new record types take the place of the normal Text records in the object program.) The three parts of the object program will be loaded into separate areas of memory, as illustrated above.

Describe how the assembler could separate the object program TI, TV and TC records as described above. What will be advantages and disadvantages of such type of object program.

OR

(b) Consider an SIC/XE program mentioned below:

Line	Loc	Source Statement		
5	0000	COPY	START	0
10	0000	FIRST	STL	RETADR
.				
.				
40	0014		J	CLOOP
45	0017	ENDFIL	LDA	=C'EOF'
.				
93		LTORG		
	002D	*	=C'EOF'	
.				
135	1040	RLOOP	TD	INPUT
145	1046		RD	INPUT
.				
185	105C	INPUT	BYTE	X'F1'
.				
215	1062	WLOOP	TD	=X'05'
.				
230	106B		WD	=X'05'
.				
255			END	FIRST
	1076		=X'05'	

Suppose we made the following changes to the program

i. Delete the LTORG statement on line 93.
ii. Change the statement on line 45 to +LDA.
iii. Change the operands on lines 135 and 145 to use literals (and delete the line 185)

Show the resulting object code for lines 45, 135, 145, 215 and 230. Also show the literal pool with address and data values.

The opcodes of the Mnemonics are as STL - 14, J – 3C, LDA- 00, TD – E0, RD – D8, WD – DC.